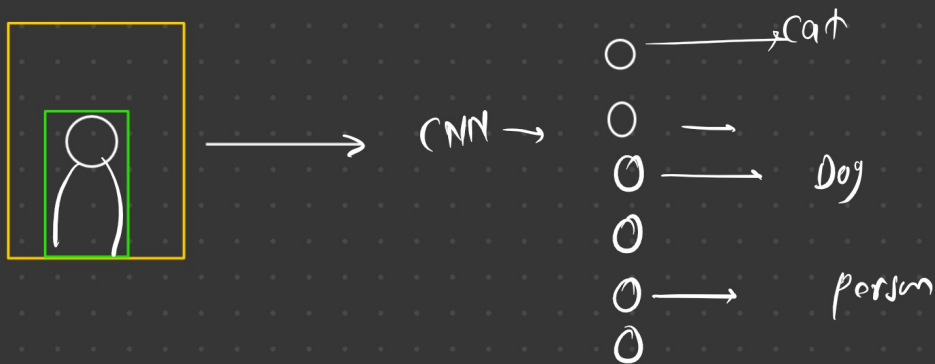


Agenda 12-01-2025

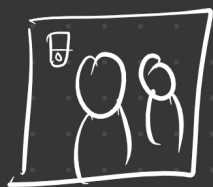
Object Detection —

- 2 stage detectors
- RNN
- Fast RNN
- Faster RNN

Image classification



Object localization



→ class!
→ bbox!

x, y x', y'

→ class
→ bbox of an object

problem statement

- multiple instance of object
- multiple classes

$$\rightarrow \text{classes} = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.7 \\ 0.8 \\ 0.9 \end{bmatrix}$$

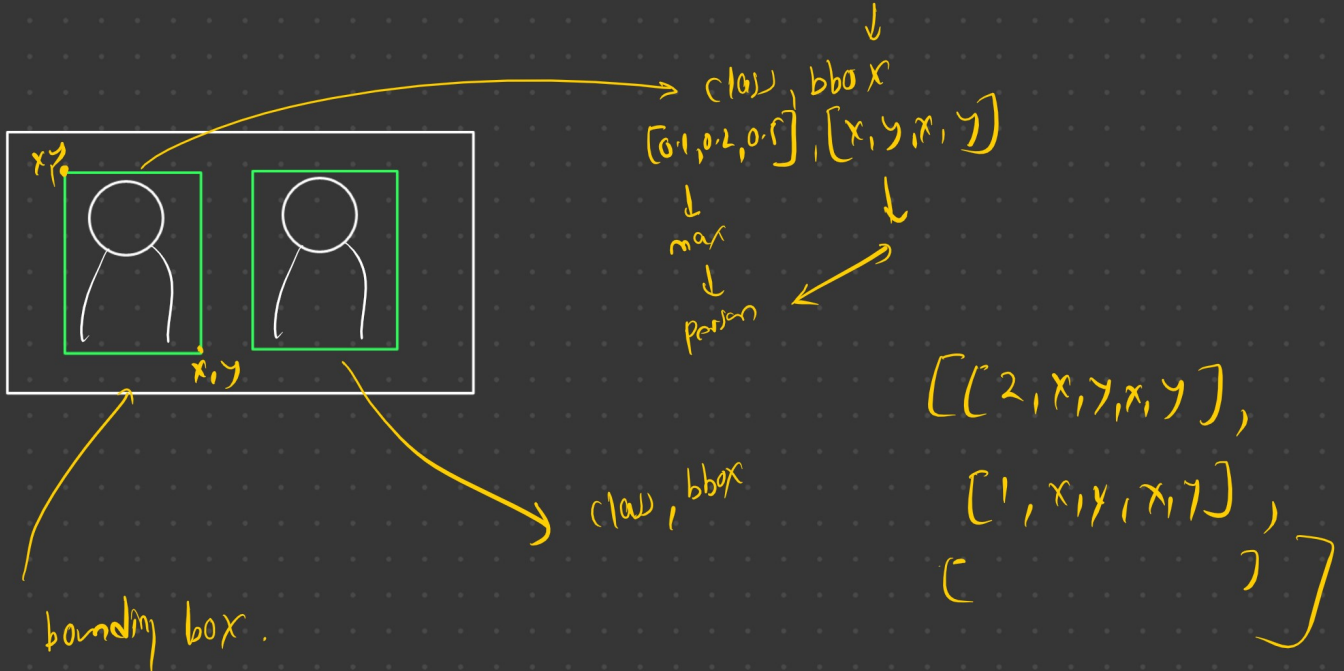
Object detection

$$\rightarrow \text{bb of regression} \rightarrow \begin{bmatrix} x \\ y \\ x \\ y \end{bmatrix}$$

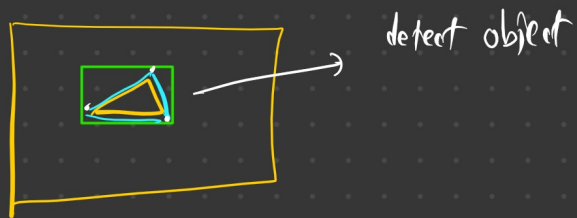
Continuous numbers

9999

$$[0, x, y, x, y]$$



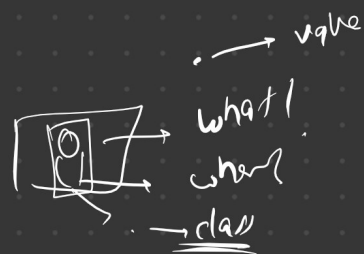
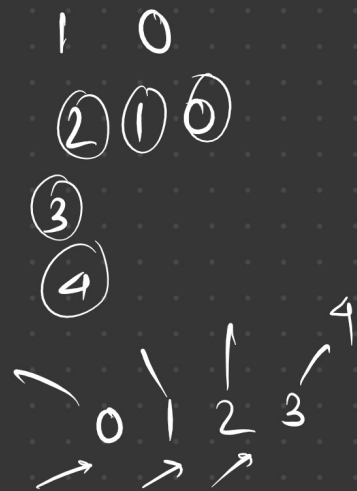
Contours →



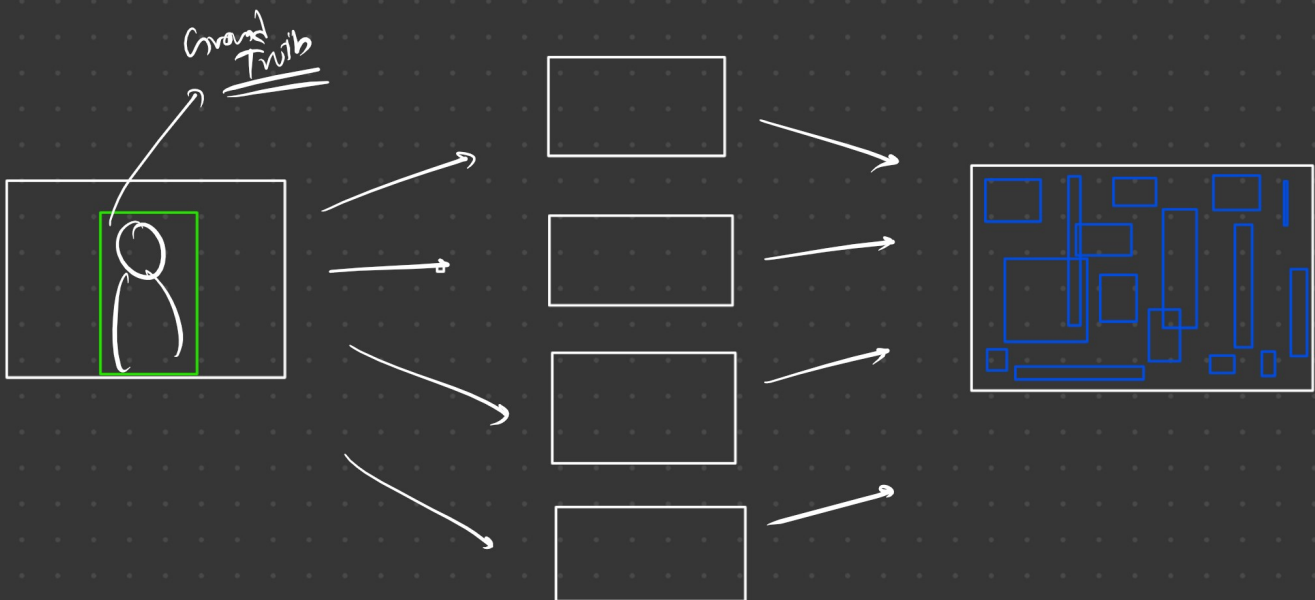
→ Draw boundary on this image

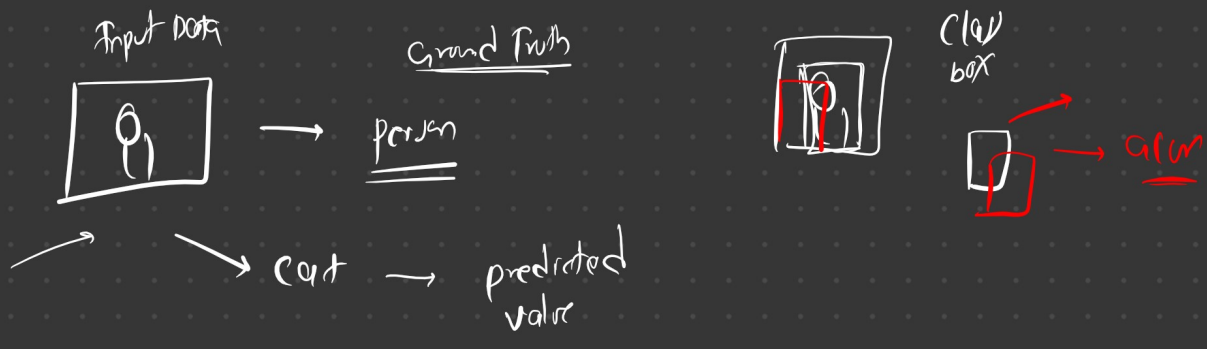
→ $[(x,y), (x,y)]$

→ cv2.



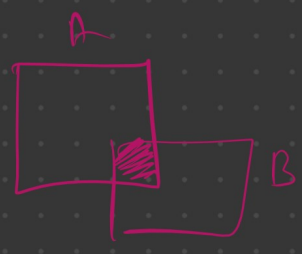
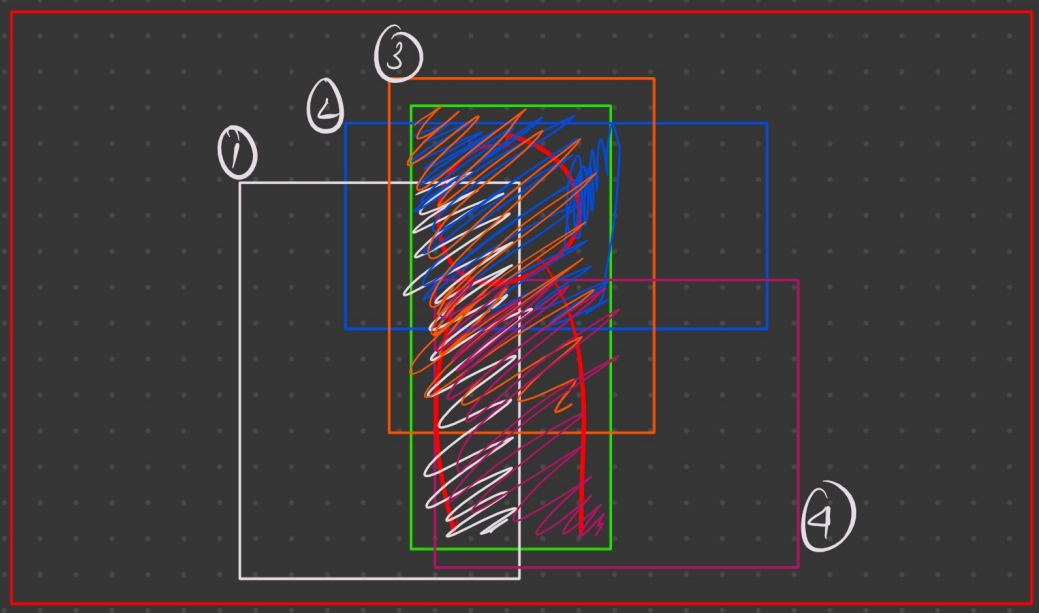
Selective



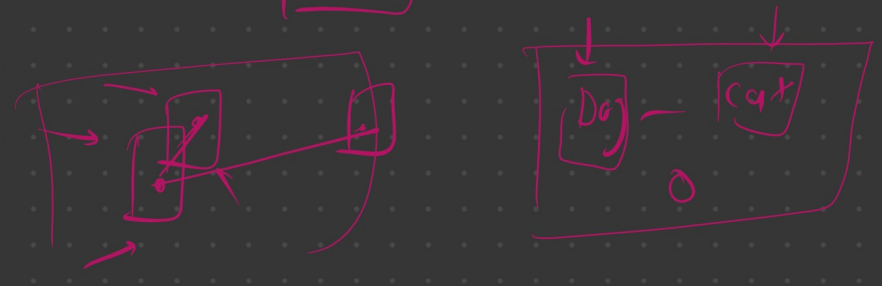


11 < 70%
→ remove

40%
45%
→ 70%
60%
700



$$\underline{FOU} = \frac{\text{Positive}}{\text{Total Area}} = 10\%$$





R-CNN : Region based CNN

2014 by Ross Girshick et al.

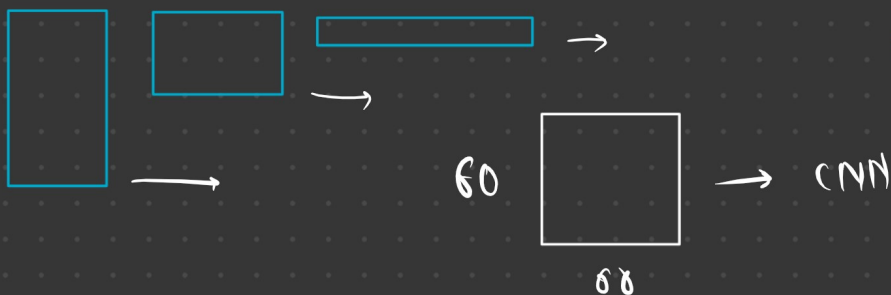
perform object detection by combining region proposals with DL.

Region proposal generation

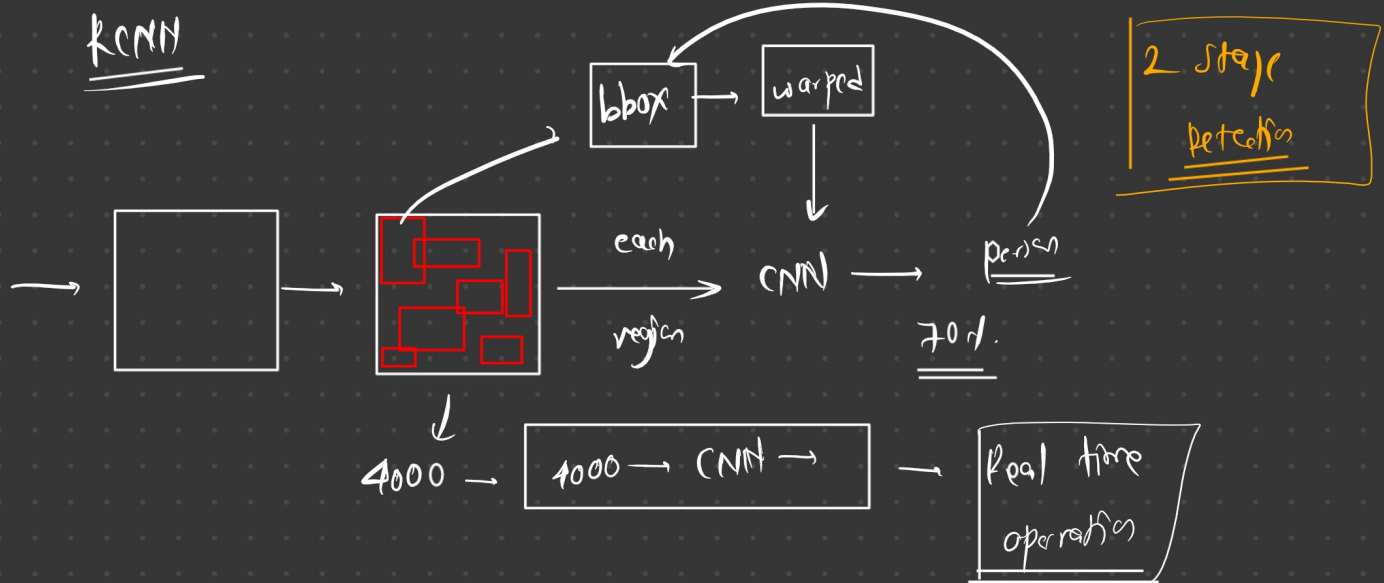
Selective Search

Image $\rightarrow$ SS $\rightarrow$ ~ 2000 region proposals $\rightarrow$

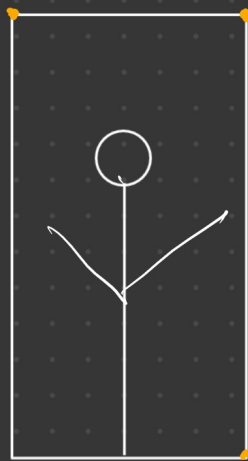
Each proposal
is a potential
object.



R-CNN

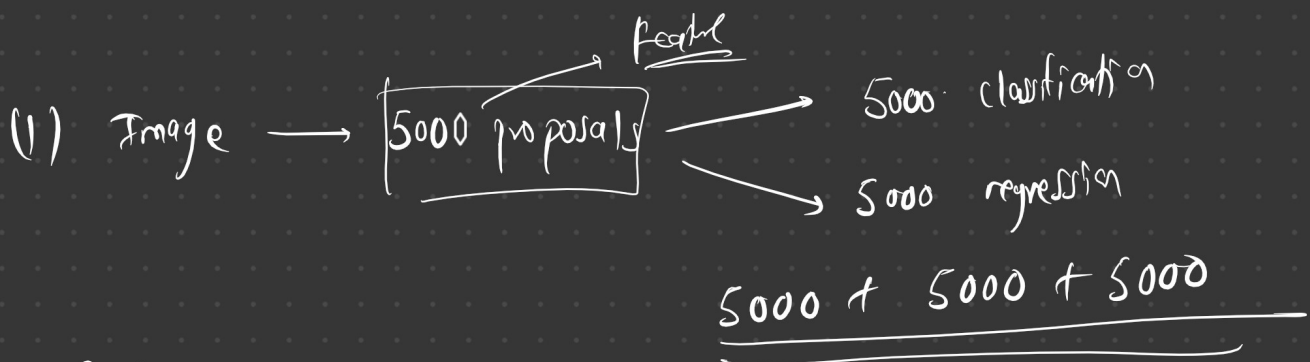


affine →



from deformation
matrix

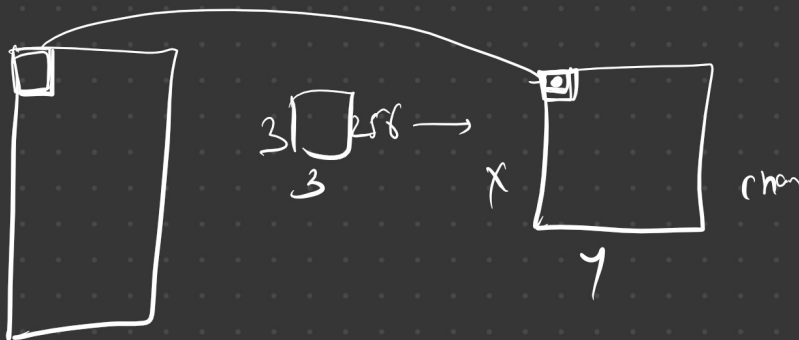
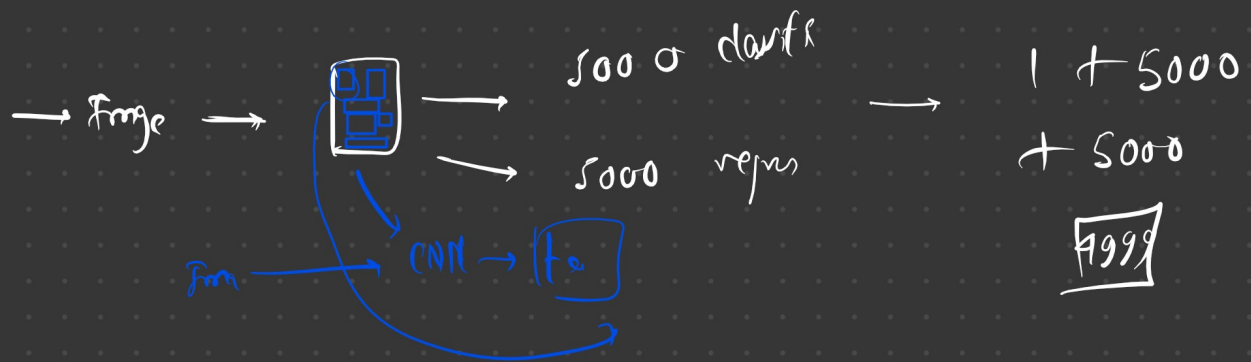
shrink



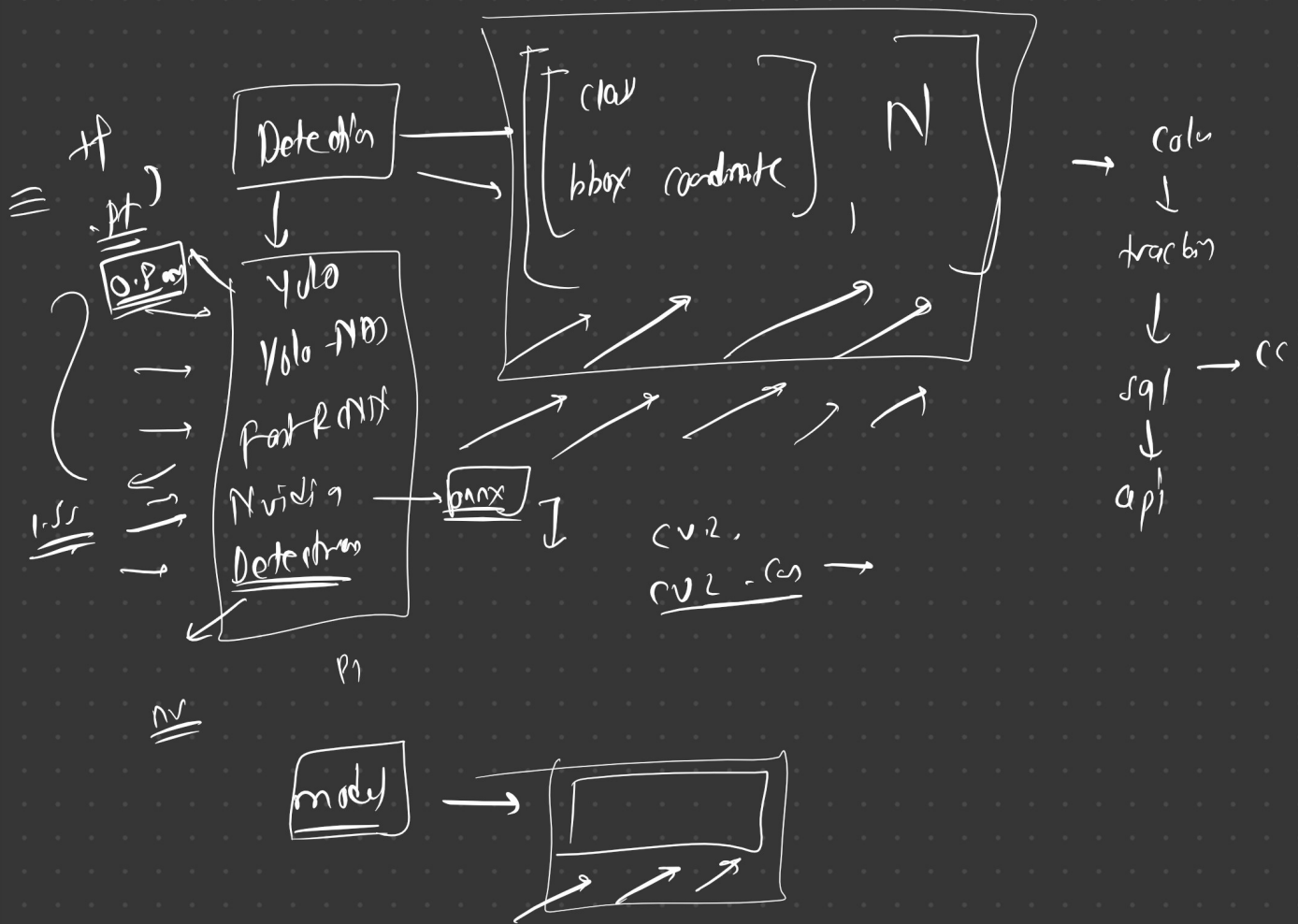
Fast R-CNN

2015, Ross Girshick

Improve R-CNN's speed & efficiency by processing the entire image in one forward pass.



Fast R-CNN



Fast RNN

2016,

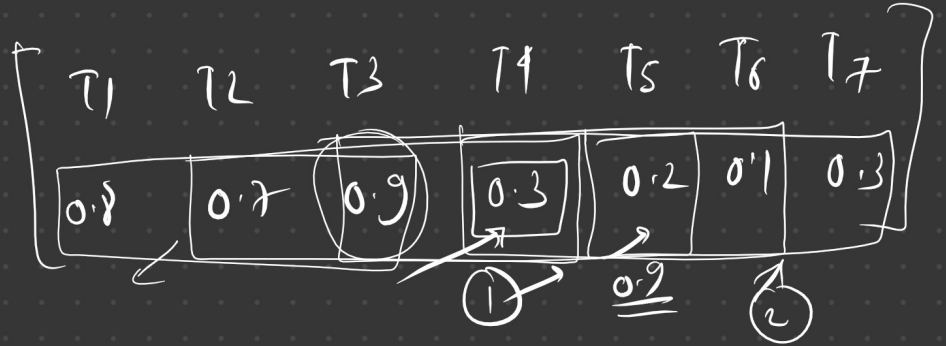
Image $\rightarrow$ CNN $\rightarrow$ feature maps $\rightarrow$ RPN
 $\downarrow$
 yyy

RPN →



feature map

4 → qyt



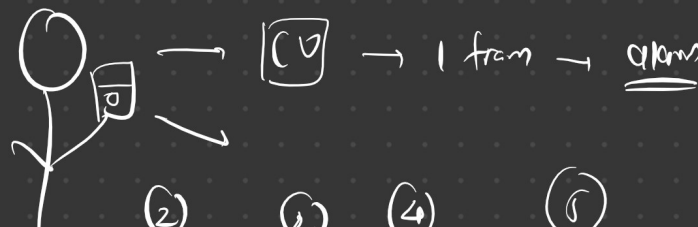
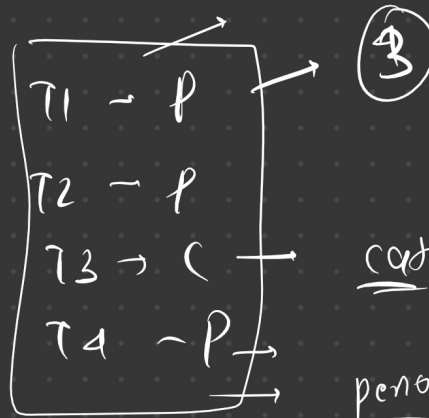
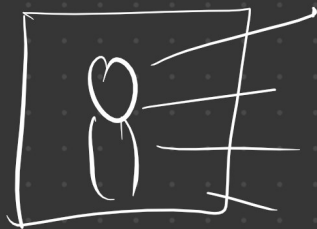
final, there is no object in the video!

→ no object

(1) → list

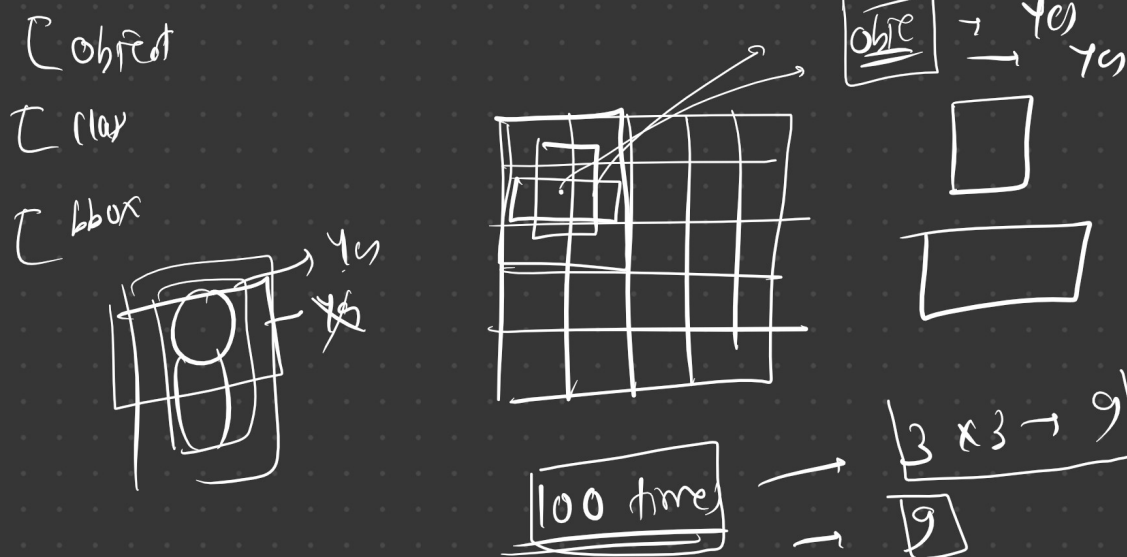
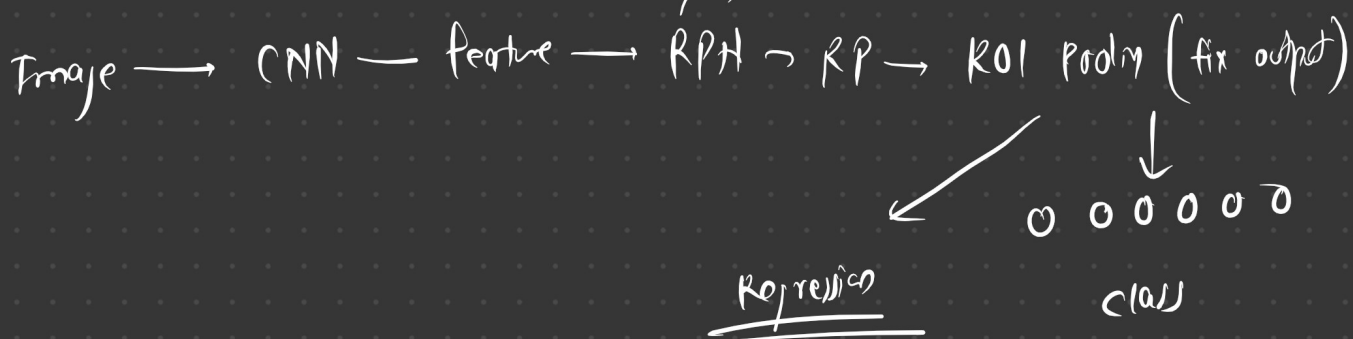
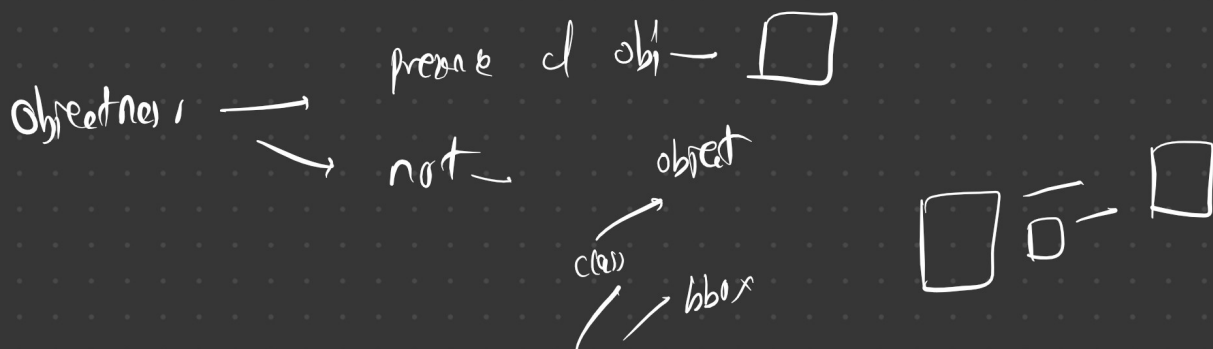
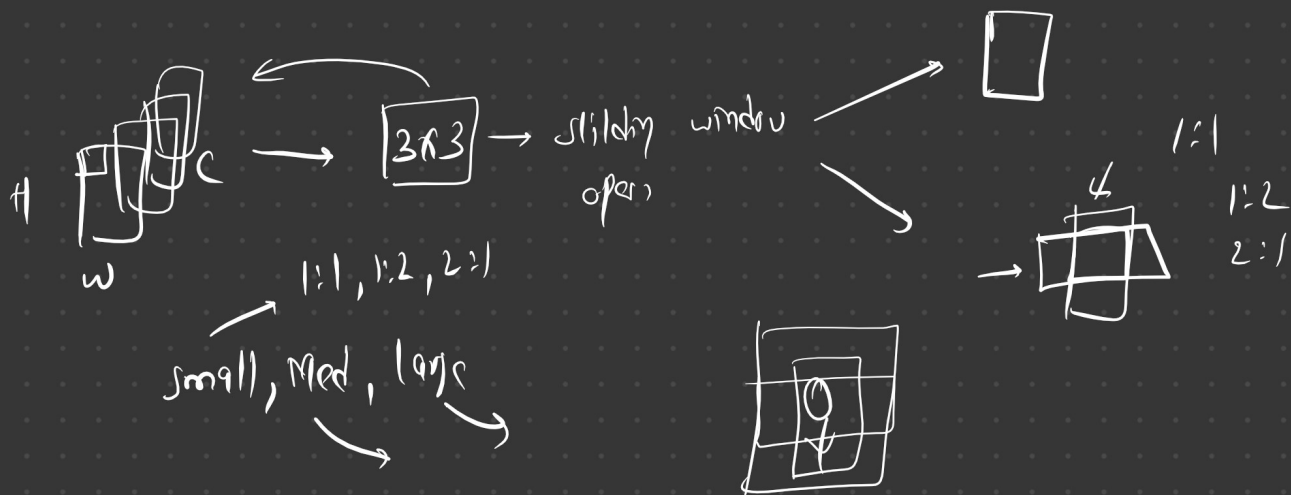
(2) → running window
→ [3]

2 times →



② T-1, ③ T-2, ④ T-3, ⑤ T-4, ⑥ T

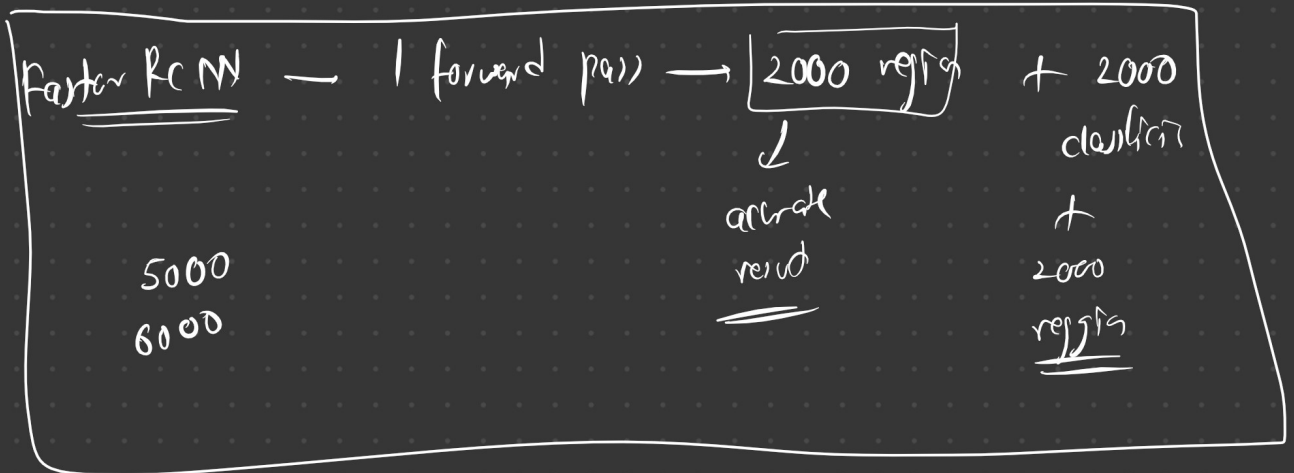
⑤ →



selectors →

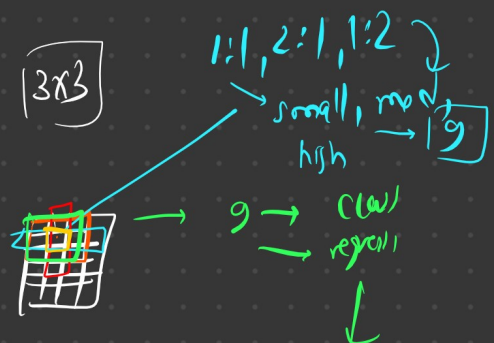
RPN $\rightarrow$ learning network

faster RNN —



7810

7 day -



ROI Pooling

14x11

7x7

00000000 ←

